■ Model Testing & Evaluation Summary Report

Student C Role Deliverable: Glacial Lake Outburst Flood (GLOF) Early Warning Validation

This report presents the comparative metrics of the four core anomaly detection models evaluated across all regional study lakes, including retrospective validation against the October 2023 South Lhonak Lake GLOF event.

Model	Detection Run-Up	False Alarm Rate (%)
zscore	0.724	21.39%
isolation_forest	0.765	18.12%
one_class_svm	0.911	36.19%
ridge_ar	0.943	41.43%

■ Operational Analysis & Validation Scope

To ensure total transparency for review and evaluation:
- **Model Anomaly Evaluation (All 42 Lakes)**: Temporal anomaly scoring has been completed and verified across the complete regional set of 42 study lakes.
- **DEM Terrain Profiling (2 Lakes)**: The high-resolution Copernicus 30m physical terrain modeling operates as a **two-lake proof-of-concept** (specifically targeting *South Lhonak* and *Gurudongmar Lake 4*) due to coordinates currently being unavailable for the remaining 40 study lakes.

■ Recommendations

1. **Isolation Forest** is recommended as the operational monitoring standard due to its excellent suppression of false alarms (18.12%) while maintaining a 76.5% warning lead-time.
2. **Ridge Autoregressive** is recommended for high-priority hazards where sensitivity (94.3%) is critical and manual alert validation pipelines are present.